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Implementation of echo client server using TCP/UDP sockets

AIM:

TO implement an Echo client server application using TCP/UDP sockets and send a message from the client to the server and receive the same message back.

ALGORITHM:-

TCP Echo server:

* Start the TCP server
* Create a serverSocket and wait for the client
* Accept the client connection
* Read the message sent by the client
* Send the same message back to the client
* Display the echoed message
* Close the connection

UDP Echo Server:

* Start the UDP server
* Create a Datagram Socket
* Receive the message from the packet
* Send the same message back to the client.
* Repeat until bye is received
* close the socket.

UDP Echo client:

* Start the UDP client
* Create a DatagramSocket
* Read a message from user.
* Create a datagram packet and send it to server
* Receive the echoed packet from server

OUTPUT :

TCP Echo

Server :

TCP Echo server started

Recived : Hello

Recived : welcome

Recived : bye

Client :

Connected to TCP Echo Server

Enter message : Hello

Server : Echo : Hello

Enter message : Welcome

Server : Echo : welcome

Enter message : bye

Server : Echo : bye

Tcp Echo client:

* Start the TCP Client
* Create a Socket and connect to the server.
* Read a message from the user.
* Send the message to the server.
* Receive the echoed message from the server.
* Display the echoed message.
* Close the connection.

CODE:

Tcp Echo server

```
import java.io.*;
import java.net.*;
public class TCPEchoServer {
    public static void main(String[] args) throws IOException {
        ServerSocket server = new ServerSocket(5000);
        System.out.println("Tcp Echo Server Started");
        Socket socket = server.accept();
        BufferedReader in = new BufferedReader(
            new InputStreamReader(socket.getInputStream()));
        PrintWriter out = new PrintWriter(
            socket.getOutputStream(), true);
        String message;
        while ((message = in.readLine()) != null) {
            System.out.println("Received: " + message);
            out.println("Echo: " + message);
            if (message.equalsIgnoreCase("bye"))
                break;
        }
        socket.close()
        server.close()
    }
}
```

Tcp Echo client

```
import java.io.*;
import java.net.*;
public class TcpEchoClient {
  public static void main (String [] args) throws IOException {
    Socket socket = new Socket ("localhost", 5000);
    printwriter out = new printwriter (
      socket.getoutputstream (), true);
    BufferedReader keyboard = new BufferedReader (
      new InputStreamReader (System.in);
    System.out.println ("connected to TCP Echo Server");
    String message;
    while (true) {
      System.out.print ("Enter a message:");
      if (message.equalsIgnorecase ("bye"))
        break;
    }
    socket.close ();
  }
}
```

UDP Echo Server

```
import java.net.*;
public class UDPEchoServer {
  public static void main (String [] args) throws Exceptio
  {
    DatagramSocket socket = new DatagramSocket (5000);
    System.out.println ("UDP Echo Server Started");
    byte[] buffer = new byte [1024];
    while (true) {
      DatagramPacket packet = new DatagramPacket ();
      socket.receive (packet);
      String message = new String (packet.getData (), 0);
      System.out.println ("Received: " + message);
      String response = "Echo :" + message;
```

OUTPUT :-

Server:

UDP Echo Server Started

Received : Hello

Received : Welcome

Received : bye

Client:

Connected to UDP Echo Server

Enter message : Hello

Server : Echo : Hello

Enter message : welcome

Server : Echo : Welcome

Enter message : bye

Server : Echo : bye

```
        if (message.equalsIgnoreCase("bye"))
            break;
    }
    socket.close();
  }
}
```

UDP Echo client

```
import java.net.*;
import java.io.*;
public class UDPEchoclient {
    public static void main(String[] args) throws Exception{
        DatagramSocket socket = new DatagramSocket();
        InetAddress address = InetAddress.getByName("
                                                localhost");
        System.out.println("connected to UDP Echo server");
        while(true){
            System.out.print("Enter message:");
            String message = keyboard.readline();
            byte[] data = message.getBytes();
            socket.send(packet);
            byte[] buffer = new byte[1024];
            socket.receive(reply);
            System.out.println("Server:" + message);
            if (message.equalsIgnoreCase("bye"))
                break;
        }
        socket.close();
    }
}
```

RESULT:

Thus, the Echo Client-Server application was successfully implemented using TCP and UDP Sockets and the message sent by the client were successfully echoed by the server.

Bill 17/9